Who Stays, Who Leaves: A Predictive Model for Employee Attrition

ECON 3916 - Machine Learning for Economics | Dat Do | Northeastern University

This report presents a machine learning analysis of employee attrition using the IBM HR Analytics Employee Attrition dataset. The goal is to move from a clear prediction question to a practical HR recommendation: identifying employees or employee segments with elevated predicted attrition risk so that managers can prioritize proactive, non-punitive retention outreach. The report summarizes the data, modeling approach, results, uncertainty, and final recommendation while emphasizing that the model is predictive rather than causal.

1. Problem Statement:

a. **Prediction question:** Can we predict whether an employee will voluntarily leave the organization based on their:
   i. demographic,
   ii. compensation, and
   iii. job-satisfaction characteristics?

b. **Stakeholder:** An HR director at a mid-size firm deciding which employee segments to target with retention programs (flexible scheduling, compensation adjustments, or career development), prioritizing segments where predicted attrition risk is highest, and intervention cost is lowest.

c. **Decision it enables:** The model produces an individual-level attrition risk score that HR can use to prioritize outreach conversations before employees decide to leave. Rather than reacting to resignations, the firm can allocate retention resources proactively toward the segments most likely to churn.

d. **Prediction vs. causation:** This is a prediction task, not causal inference. The model identifies which features are associated with higher attrition rates in this dataset, but it does not establish that any feature causes attrition. For example, OverTime is a strong predictor, but that does not mean eliminating overtime would necessarily reduce attrition. Confounders such as job role, workload, and compensation may drive both overtime exposure and departure decisions simultaneously. All feature importance results in this report should be interpreted as predictive associations, not policy proof.

2. Data Description:

This project uses the IBM HR Analytics Employee Attrition dataset from Kaggle (https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset).

The dataset contains 1,470 employee records and 35 columns, 34 predictors plus the binary target Attrition (Yes/No). Features span demographics (Age, Gender, MaritalStatus), compensation (MonthlyIncome, DailyRate, StockOptionLevel), job characteristics (JobRole,

Department, JobLevel), satisfaction scores (JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance), and tenure (YearsAtCompany, TotalWorkingYears, YearsWithCurrManager).

a. **Quality assessment:** The dataset is complete with no missing values, MCAR/MAR/MNAR classification is not applicable, and no imputation was required. Four constant or noninformative columns were dropped: EmployeeCount, StandardHours, Over18, and EmployeeNumber.

b. **Class imbalance:** Only 16.1% of employees left (n=237) while 83.9% stayed (n=1,233). This imbalance means that a naive classifier predicting "Stayed" for all employees would achieve 84% accuracy while identifying zero true leavers. For this reason, recall and F1-score for the "Left" class are treated as the primary evaluation criteria throughout this analysis.

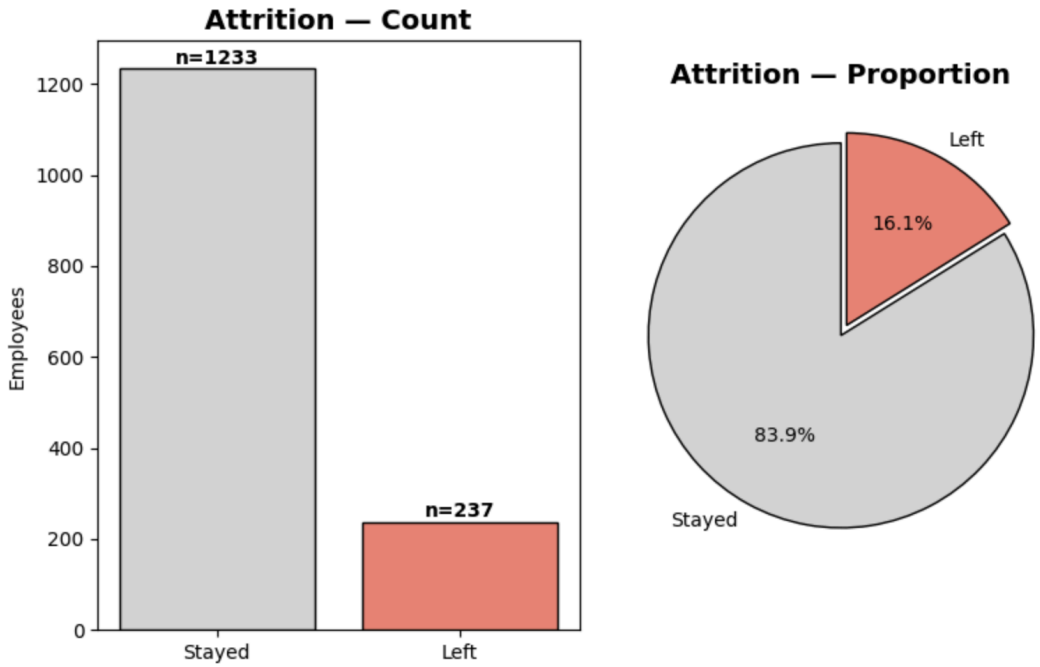

Figure 1: Class imbalance

c. **Key EDA findings:** Employees who work overtime have an attrition rate of 30.5% vs. 10.4% for non-overtime workers, a threefold gap. Employees who left have a lower median monthly income ($3,202) compared to those who stayed ($5,204). The correlation heatmap shows that TotalWorkingYears, JobLevel, YearsInCurrentRole, MonthlyIncome, and Age are the strongest negative correlates of attrition, suggesting that earlier-career, lower-paid, and younger employees are more likely to leave. Individual correlations are modest (max |r| = 0.17), indicating the model must combine multiple weak signals rather than rely on a single dominant predictor.

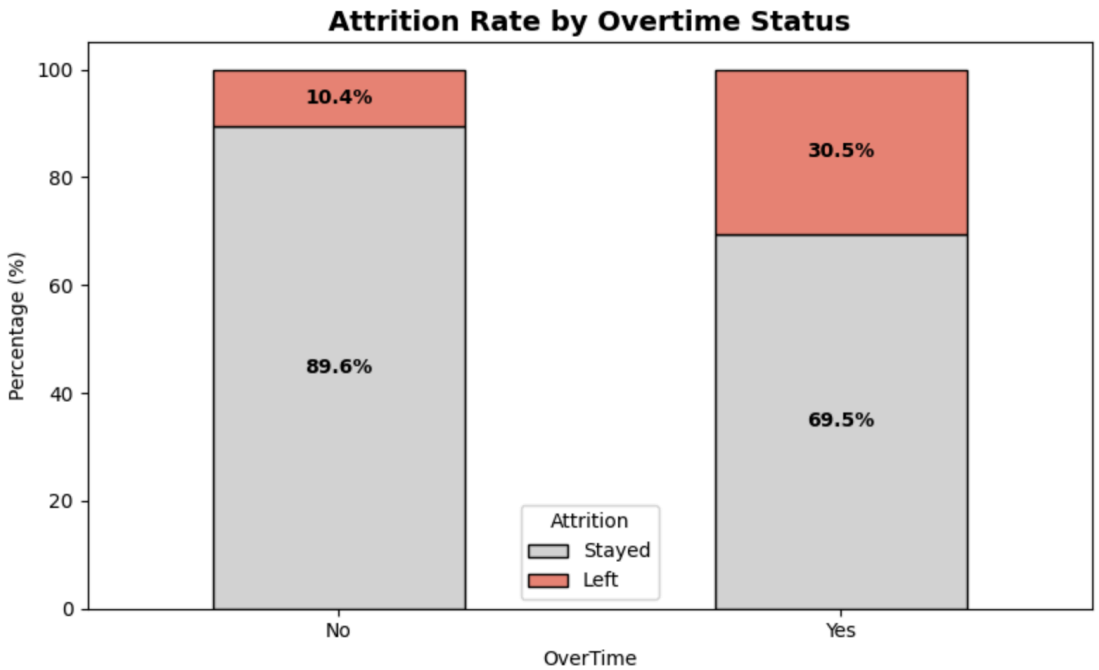

Figure 2: Attrition Rate by Overtime Status

d. **Outlier assessment (IQR method):** Upper-tail outliers were detected in MonthlyIncome (7.8%), YearsAtCompany (7.1%), TotalWorkingYears (4.3%), and NumCompaniesWorked (3.5%). These reflect genuine workforce heterogeneity, senior executives and frequent job-changers, rather than data errors. All records were retained.

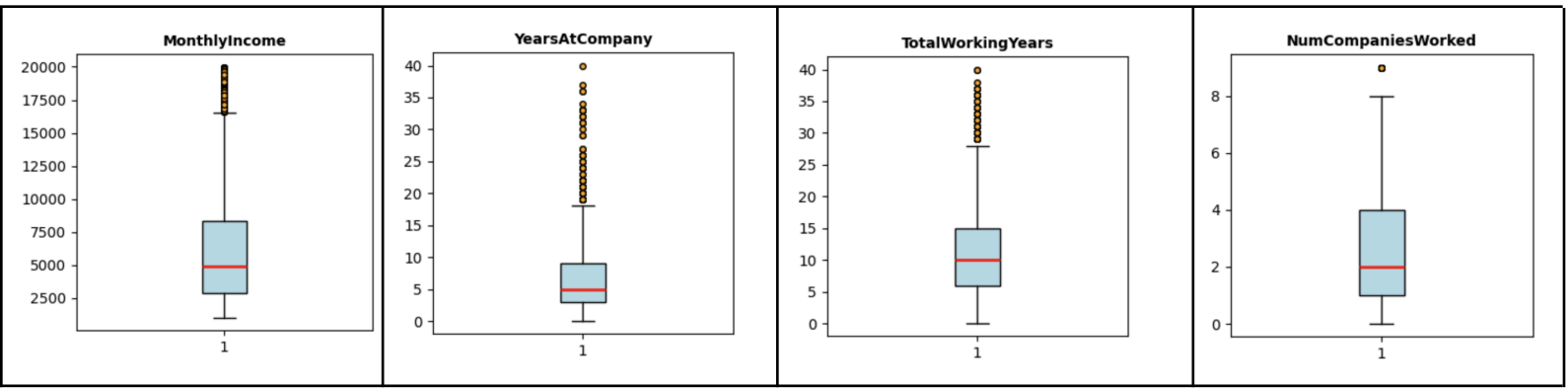

Figure 3: Boxplot of MonthlyIncome, YearsAtCompany, TotalWorkingYears, NumCompaniesWorked

3. Methodology

a. **Preprocessing:** Categorical features were one-hot encoded with drop_first=True, producing 44 predictors after encoding. Features were standardized using StandardScaler fit on the training set only, with the fitted scaler applied to the test set to prevent data leakage.

b. **Train/test split:** The dataset was split 80/20 with random_state=42 and stratify=y to preserve the 16% attrition rate in both sets. The training set contains 1,176 samples (16.2% attrition) and the test set contains 294 samples (16.0% attrition).

c. **Models compared:** Two models were evaluated.
   i. The first is Logistic Regression, which is a simple, interpretable, and widely used as a binary classification baseline.
   ii. The second is Random Forest with 200 trees, a nonlinear ensemble method that can capture feature interactions that logistic regression may miss.

   No heavy hyperparameter tuning was performed; the goal was a transparent comparison between a simple parametric model and a more flexible alternative.

d. **Cross-validation:** 5-fold stratified cross-validation was used to evaluate generalization performance. For Logistic Regression, a pipeline that wrapped StandardScaler within each fold was used to prevent data leakage. The scaler was refit on each fold's training data separately. F1-score was the primary CV metric given the class imbalance.

e. **Confidence intervals:** Bootstrap 95% confidence intervals were computed using 1,000 resamples of the held-out test set for accuracy, precision, recall, F1, and AUC. A separate bootstrap function with a single-class guard was used for AUC to handle bootstrap samples where only one class appears.

4. Results

Table 1: Test set performance with 95% bootstrap confidence intervals

Metric	LR Mean	LR 95% CI	RF Mean	RF 95% CI
Accuracy	0.8605	[0.8230, 0.8980]	0.8265	[0.7857, 0.8707]
Precision	0.6154	[0.4091, 0.8000]	0.3333	[0.0769, 0.6364]
Recall	0.3404	[0.2036, 0.4773]	0.0851	[0.0189, 0.1750]
F1	0.4384	[0.2856, 0.5714]	0.1356	[0.0322, 0.2667]
AUC	0.8079	[0.7301, 0.8791]	0.8009	[0.7310, 0.8673]

Table 2: 5-fold stratified cross-validation F1 scores on training set

Model	Mean CV F1	Std	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Logistic Regression	0.5552	0.0884	0.5263	0.4000	0.6129	0.6562	0.5806
Random Forest	0.3044	0.0834	0.2326	0.1860	0.4167	0.3404	0.3462

Logistic Regression outperforms Random Forest across all five test set metrics and in every cross-validation fold. The most important gap is in recall, Logistic Regression identifies 16 of the 47 actual leavers in the test set (34.0%) while Random Forest identifies only 4 (8.5%) (see highlighted).

The non-overlapping 95% bootstrap confidence intervals for recall and F1 suggest that Logistic Regression's advantage on these metrics is unlikely to be due only to test-set sampling variation. AUC confidence intervals overlap slightly (LR [0.73, 0.88] vs RF [0.73, 0.87]), so the strongest evidence for Logistic Regression comes from recall, F1, and cross-validated F1 rather than a large difference in ranking performance.

Feature importance (predictive, not causal):

The top positive predictors of attrition risk from the Logistic Regression standardized coefficients are OverTime_Yes (0.865), BusinessTravel_Travel_Frequently (0.751), and JobRole_Laboratory Technician (0.715).

The top negative predictors are TotalWorkingYears (-0.556), EnvironmentSatisfaction (-0.482), and JobSatisfaction (-0.419).

These reflect predictive associations within this model. A high coefficient for OverTime means overtime is a useful signal for predicting attrition, it does not prove that eliminating overtime would reduce departures.

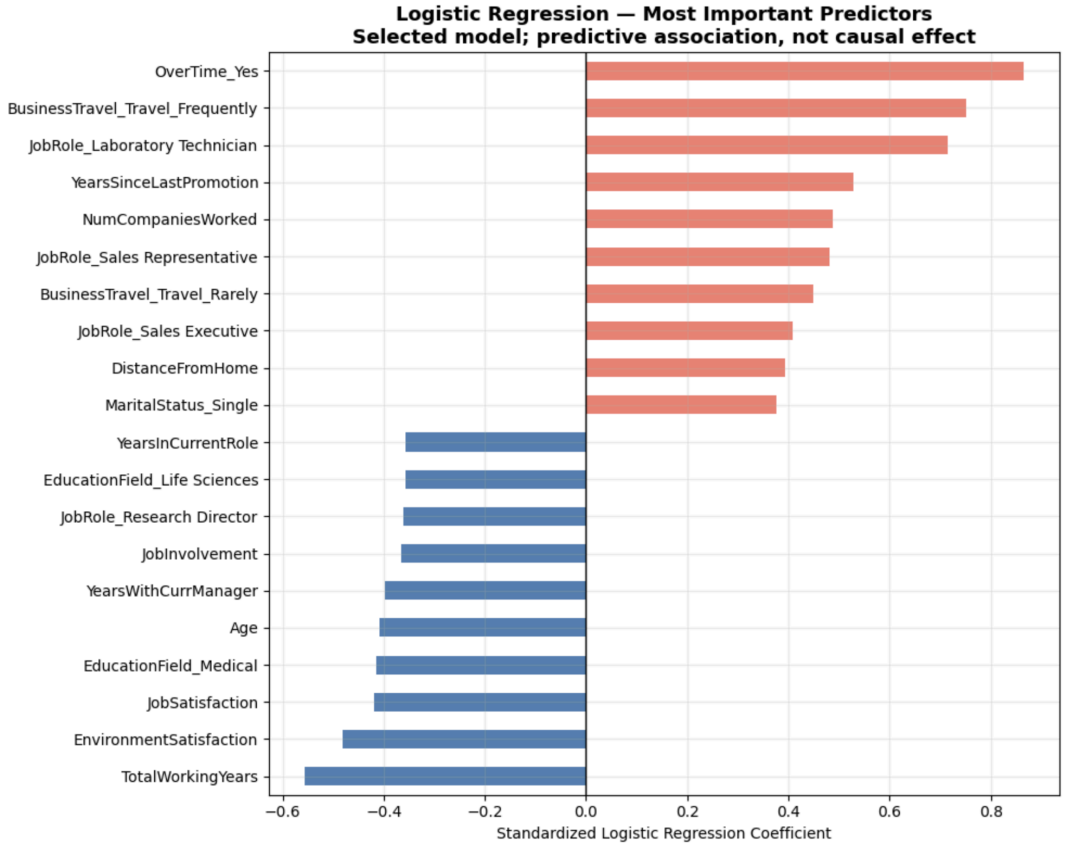

Figure 4: Logistic Regression: Most Important Predictor
Predictive importance, not causal effect. Feature coefficients reflect model weights, not policy levers.

5. Recommendation

a. **Actionable recommendation:** Pilot the Logistic Regression model as an HR risk-screening tool. In a pilot setting, HR could score active employees monthly and use the risk scores to prioritize voluntary, non-punitive outreach conversations. The model should support human judgment rather than make automatic employment decisions.

b. **Priority segments:** Based on the coefficient analysis, HR should prioritize employees who work overtime, travel frequently, are in Laboratory Technician or Sales Representative roles, are single, and have not received a promotion recently. These groups combine multiple positive predictors and represent the highest-risk segments in the model.

c. **Uncertainty bounds:** The 95% bootstrap confidence interval for Logistic Regression F1 is [0.29, 0.57], reflecting meaningful uncertainty. The model's recall of 34.0% means it misses roughly two-thirds of actual leavers at the default 0.50 threshold. HR teams should understand that the model captures only a minority of true attrition cases, not a comprehensive signal.
Individual-level uncertainty should also be communicated clearly. A predicted attrition probability should not be read as certainty that a specific employee will leave. It is better interpreted as a prioritization score: employees with higher scores share characteristics that, in this dataset, are associated with higher observed attrition. For that reason, the model is most appropriate for broad screening and resource allocation, not for labeling employees as disloyal or making punitive decisions.

d. **Threshold recommendation:** A practical next step is to evaluate lower classification thresholds, such as 0.25–0.30, on a validation set. Lower thresholds would generally increase recall, though at the cost of more false positives. Because the proposed interventions are low-cost and non-punitive, this tradeoff may be acceptable, but the exact threshold should be chosen using the firm's intervention capacity and cost structure.

e. **Limitations:** The model was trained on a single IBM HR dataset with 1,470 employees. It may not generalize to firms with different industries, sizes, or compensation structures. The 16% attrition rate and feature distributions are specific to this sample. Model performance should be re-evaluated after deployment on real employee data, and the model should be retrained periodically as workforce composition changes. The model also depends on the variables available in the IBM dataset. Important drivers of turnover, such as manager quality, external job offers, commute conditions, team culture, and recent life changes, are not directly observed. Because these omitted factors may influence both the observed predictors and attrition outcomes, the model should not be interpreted as a complete explanation of why employees leave.

f. **Next steps:** First, evaluate lower classification thresholds on a validation set to find the optimal recall-precision tradeoff for the specific intervention cost structure. Second, collect firm-specific labeled data and retrain the model to improve generalization. Third, conduct exit interviews to validate whether model-flagged employees share the risk factors the model identifies, providing a partial check on predictive validity before deployment.

6. Access Link:

1. Live dashboard: https://finalproject-datdo-econ3916.streamlit.app/

2. GitHub: https://github.com/ddoneu/FinalProject-Employee-Attrition-Prediction

3. Colab: https://colab.research.google.com/drive/1ROb8NA4XWTTqnuxHvFYjI0P74kPOUYcS?authuser=2#scrollTo=9y6iJRBz3Kwj